LAB 19 - PROMPTS

Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab19-performance-report/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Make sure the forward-test logs exist (A8)

Do this (no prompt needed)
The stage reads _outputs/Lab16_equity_log.csv and _outputs/trade_log.csv from the Lab 16 runner; if either is missing it runs the 2025 replay itself. Run python -m pipeline run --loop if you want fresh logs.

Check: _outputs/Lab16_equity_log.csv has 250 rows ending at USD 116,249.79.

Step 2: Build the backtest-vs-forward comparison (A8)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/report.py. load_forward_test(cfg) reads the equity log and the BUY/SELL rows of trade_log.csv whose run_time falls in 2025-01-02 to 2025-12-31 (running pipeline.runner.replay quietly if a log is missing). In run(): backtest the configured strategy 2018-2025 with the config cost model and print a metrics table versus SPY buy and hold (total return, CAGR, volatility, Sharpe, Sortino, max drawdown, Calmar, win rate, profit factor, trades, exposure, alpha, beta) and the cost breakdown; backtest 2025 only (starting from 2024-12-31 so the first decision is warm) and compare it with the forward test: total return, Sharpe, max drawdown, orders and average fill slippage in bps versus the day's open; print the tracking difference and annualised tracking error. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: The comparison table has the columns 'Backtest 2025' and 'Forward test 2025 (paper)'.

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Step 3: Add monitoring checks, the tearsheet and the report (A8, A10)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/report.py add THRESHOLDS = {'tracking_abs': 0.02, 'slippage_bps': 5.0, 'drawdown_stop': 0.15} and monitoring checks with status OK or ALERT: tracking vs backtest, average fill slippage, drawdown stop rule (current and worst forward-test drawdown), kill switch (pipeline.risk.RiskManager) and diagnostics (read _outputs/Lab18_diagnostics.csv rows for the configured strategy: ALERT on any FAIL or if diagnose was not run for it). Save _outputs/Lab19_monitoring.csv; a 3 x 2 tearsheet Lab19_tearsheet.png (equity vs SPY with the forward-test year shaded, drawdowns, monthly-returns heatmap fixed at +/-8%, rolling 6-month Sharpe, backtest vs forward 2025, costs by type); and _outputs/Lab19_report.md with the tables (DataFrame.to_markdown), five data-supported insights and a recommendation: 'CONTINUE PAPER TRADING ...' if every check is OK, otherwise 'PAUSE AND INVESTIGATE: monitoring alerts on ...'. Paper results only - not investment advice. main(argv) for `python -m pipeline report`. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline report --help works.

Step 4: Generate the report (A8, A10)

Do this (no prompt needed)
Run: python -m pipeline diagnose (so diagnostics exist for ema_cross), then python -m pipeline report (same as: python3 Lab19-performance-report/report_stage.py).

Check: 2018-2025: strategy CAGR 9.02% vs SPY 14.15%, Sharpe 0.81 vs 0.78, max drawdown -14.19% vs -33.72%, beta 0.35, alpha +4.03%; costs USD 1,072 (drag 0.11% a year). 2025: backtest +17.04% (Sharpe 1.64) vs forward test +16.25% (Sharpe 1.65), 5 orders each, average slippage 1.0 bps, tracking difference -0.79%.

Step 5: Read the monitoring checks (A8)

Do this (no prompt needed)
Read section [3]. Which limit is closest to being breached?

Check: Five checks OK: tracking -0.79% (limit +/-2%), slippage 1.0 bps (limit 5), drawdown current -1.34% / worst -4.85% (stop at -15%), kill switch off, diagnostics '1 WARN, no FAIL'.

Step 6: Trigger an alert on purpose (A8)

Do this (no prompt needed)
In pipeline/report.py change THRESHOLDS['tracking_abs'] to 0.005 and rerun python -m pipeline report. Then restore 0.02.

Check: The tracking check shows ALERT and the recommendation becomes 'PAUSE AND INVESTIGATE: monitoring alerts on tracking vs backtest'.

Step 7: Study the tearsheet (A10)

Do this (no prompt needed)
Open _outputs/Lab19_tearsheet.png: equity, drawdown, monthly-returns heatmap, rolling Sharpe, backtest vs forward, costs. Find the weakest months.

Check: The forward-test and backtest lines for 2025 almost overlap; the heatmap shows the weak months in red.

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Step 8: Write the committee recommendation (A10)

Prompt - copy into your AI coding assistant
Read labs_fullcode/_outputs/Lab19_report.md. Rewrite sections 5 and 6 for Lion City Capital's investment committee in plain English (under 200 words): what the data show, why the forward test matters, the honest message that the strategy reduces risk but has not beaten buy and hold, one additional risk (for example regime change or capacity), and a clear, conditional recommendation. Keep every number unchanged.

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Check: A recommendation that continues paper trading with conditions (second signal, drawdown stop, risk-committee sign-off).

Run it

Step	Command / file
Run Lab 19	python3 Lab19-performance-report/report_stage.py (same as: python -m pipeline report)
The complete code	pipeline/report.py (reference copy: Lab19-performance-report/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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